

Nelson Cala-Moreno

DVM · MSC · PHD

Quantitative Genetics · Biostatistics · Genomics · Animal Science

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PROFILE

Doctor of Veterinary Medicine and PhD in Animal Production and Health Sciences (UNAM, Mexico), specialising in genetics and biostatistics applied to animal sciences. Experienced in university teaching, research leadership and the supervision of graduate work in quantitative genetics, genomics, experimental design and research methodology. Skilled in the analysis of complex biological data through mixed models, multivariate techniques and genome-wide association studies (GWAS), with a research focus on genetic characterisation, animal adaptation and health under tropical conditions.

ACADEMIC INDICATORS

57

Google Scholar citations

4

h-index

20+

Theses supervised

3+

Projects as PI

EDUCATION

PhD, Animal Production and Health Sciences

Universidad Nacional Autónoma de México · 2021

Animal Genetics and Biostatistics. CONACYT scholarship holder.

MSc, Animal Production and Health Sciences

Universidad Nacional Autónoma de México · 2014

Animal Genetics and Biostatistics. CONACYT scholarship holder.

Doctor of Veterinary Medicine (DVM)

Universidad Cooperativa de Colombia · 2005

Both graduate degrees officially recognised in Colombia by the Ministry of National Education.

Selected training

Design of Breeding Programs with Genomic Selection (University of New England); Phenotypic Prediction Workshop (University of Florida); Mixed Models with ASReml (UNAM).

PROFESSIONAL EXPERIENCE

Full-Time Research Professor · Universidad Cooperativa de Colombia

Bucaramanga, Colombia · February 2022 – June 2026

- Undergraduate and graduate teaching in genetics, quantitative genetics, biostatistics, experimental design and research methodology.
- Principal investigator on funded projects in *Tenebrio molitor* and the Santandereana Creole goat, in collaboration with UNAM and UAM Xochimilco.
- Supervision of more than 20 completed dissertations and master's theses; contribution to graduate curriculum design.
- Methodological and statistical consulting for institutional research projects.

Research Projects Consultant · Independent

January 2018 – Present

- Methodological and statistical consulting in animal genetics, animal production and veterinary sciences.
- Experimental design and data analysis using mixed models, multivariate analysis and genomic tools.
- Consultancy on genomic selection, GWAS and sustainable animal production for projects in Colombia and Mexico.

Teaching Assistant · UNAM and UAM Xochimilco

Mexico City, Mexico · 2013 – 2019

- Teaching and tutoring at undergraduate, master's and doctoral level in Animal Breeding, Quantitative Genetics and Statistics.
- Training of students in data analysis using SPSS and R.

Genetic Improvement Advisor

Colombia · December 2015 – May 2017

- Genetic evaluation of registered cattle populations using statistical and genomic tools.
- Deterministic modelling of genomic selection strategies for Colombian Simmental cattle.

SELECTED PUBLICATIONS

1. Gómez-Rojas, C. J., Martínez-Marín, G., Toledo-Alvarado, H. O., & **Cala-Moreno, N.** (2026). Genetic parameters of larval weight and length in yellow mealworm under tropical conditions: heritability and implications for breeding programs. *animal*, 101784. <https://doi.org/10.1016/j.animal.2026.101784>
2. **Cala-Moreno, N.**, Caicedo Blanco, Y. A., García Arévalo, J. L., & Toledo Alvarado, H. O. (2025). La cabra criolla santandereana: patrimonio genético y sostenibilidad productiva en el nororiente colombiano. *Spei Domus*, 21(1), 1–16. <https://doi.org/10.16925/2382-4247.2025.01.11>
3. Soto, S. G. N., Villalobos, J. M. B., **Moreno, N. C.**, Monforte, J. G. M., Galván, P. O., Arvisu, R. U., & Alvarado, H. O. T. (2023). Genome-wide association study for heat stress resistance in Brown Swiss cattle in Yucatan, Mexico. *Veterinaria México OA*, 10.
4. **Cala-Moreno, N.**, Campos-Montes, G., Caballero-Zamora, A., Berruecos-Villalobos, J., & Castillo-Juárez, H. (2021). Genotype-by-environment interaction in White shrimp associated with White Spot Disease. *Abanico Veterinario*, (11), 1–16. <https://dx.doi.org/10.21929/abavet2021.13>
5. Madalena, F. E., Toledo-Alvarado, H., & **Cala-Moreno, N.** (2019). Animals That Produce Dairy Foods: Bos indicus Breeds and Bos indicus × Bos taurus Crosses. *Reference Module in Food Science*, Elsevier. <https://doi.org/10.1016/B978-0-08-100596-5.00619-3>

Full publication list, conference output and supervised theses available in the extended CV.

RESEARCH LINES AND CURRENT PROJECTS

- **Santandereana Creole goat:** genome-wide association for heat stress tolerance and gastrointestinal parasitism; physiological, haematological and biochemical characterisation under tropical conditions.
- **Edible insects (*Tenebrio molitor*, *Zophobas morio*):** estimation of genetic parameters and heritability for production traits in genetic nuclei.
- **Applied animal breeding:** genomic selection strategies, deterministic modelling and genetic evaluation in cattle and aquaculture species.

EXPERTISE

Quantitative genetics

GWAS

Genomic selection

Mixed models

Experimental design

Multivariate analysis

Meta-analysis

Systematic reviews

Animal breeding

Scientific writing

SOFTWARE · LANGUAGES · PEER REVIEW

Software: ASReml (advanced), R (advanced), JMP (advanced), MATLAB (intermediate), SPSS (intermediate), Python (basic).

Languages: Spanish (native), English (intermediate/advanced).

Peer reviewer: *Turkish Journal of Fisheries and Aquatic Sciences* (2020); *Spei Domus* (2023).

Research group: GRICA — Animal Sciences Research Group, Universidad Cooperativa de Colombia (Colciencias Category A1).